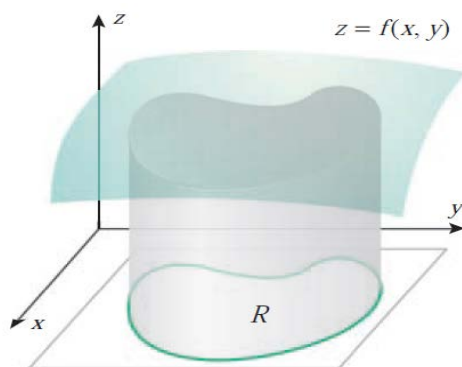


14.1 DOUBLE INTEGRALS

14.1.1 THE VOLUME PROBLEM Given a function f of two variables that is continuous and nonnegative on a region R in the xy -plane, find the volume of the solid enclosed between the surface $z = f(x, y)$ and the region R (Figure 14.1.1).

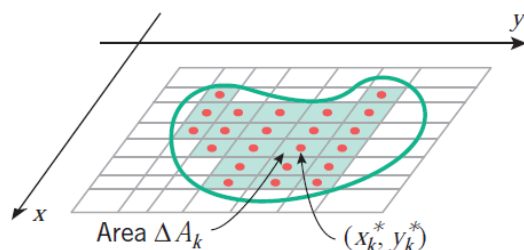


▲ Figure 14.1.1

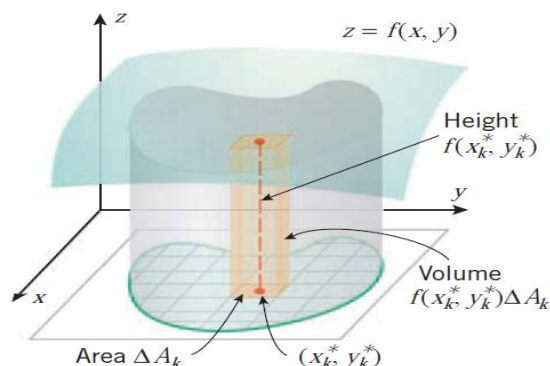
14.1.2 DEFINITION (Volume Under a Surface) If f is a function of two variables that is continuous and nonnegative on a region R in the xy -plane, then the volume of the solid enclosed between the surface $z = f(x, y)$ and the region R is defined by

$$V = \lim_{n \rightarrow +\infty} \sum_{k=1}^n f(x_k^*, y_k^*) \Delta A_k \quad (2)$$

Here, $n \rightarrow +\infty$ indicates the process of increasing the number of subrectangles of the rectangle enclosing R in such a way that both the lengths and the widths of the subrectangles approach zero.



▲ Figure 14.1.2



▲ Figure 14.1.3

■ DEFINITION OF A DOUBLE INTEGRAL

The **Riemann sums**, and the limit of the Riemann sums is denoted by

$$\iint_R f(x, y) dA = \lim_{n \rightarrow +\infty} \sum_{k=1}^n f(x_k^*, y_k^*) \Delta A_k \quad (4)$$

which is called the **double integral** of $f(x, y)$ over R .

If f is continuous and nonnegative on the region R , then the volume formula in (2) can be expressed as

$$V = \iint_R f(x, y) dA \quad (5)$$

If f has both positive and negative values on R , then a positive value for the double integral of f over R means that there is more volume above R than below, a negative value for the double integral means that there is more volume below R than above, and a value of zero means that the volume above R is the same as the volume below R .

► Example 1

$$\begin{aligned} \int_0^1 xy^2 dx &= y^2 \int_0^1 x dx = \left. \frac{y^2 x^2}{2} \right]_{x=0}^1 = \frac{y^2}{2} \\ \int_0^1 xy^2 dy &= x \int_0^1 y^2 dy = \left. \frac{xy^3}{3} \right]_{y=0}^1 = \frac{x}{3} \quad \blacktriangleleft \end{aligned}$$

$$\int_c^d \int_a^b f(x, y) dx dy = \int_c^d \left[\int_a^b f(x, y) dx \right] dy \quad (6)$$

$$\int_a^b \int_c^d f(x, y) dy dx = \int_a^b \left[\int_c^d f(x, y) dy \right] dx \quad (7)$$

These integrals are called **iterated integrals**.

► **Example 2** Evaluate

$$(a) \int_1^3 \int_2^4 (40 - 2xy) dy dx \quad (b) \int_2^4 \int_1^3 (40 - 2xy) dx dy$$

Solution (a).

$$\begin{aligned} \int_1^3 \int_2^4 (40 - 2xy) dy dx &= \int_1^3 \left[\int_2^4 (40 - 2xy) dy \right] dx \\ &= \int_1^3 (40y - xy^2) \Big|_{y=2}^4 dx \\ &= \int_1^3 [(160 - 16x) - (80 - 4x)] dx \\ &= \int_1^3 (80 - 12x) dx \\ &= (80x - 6x^2) \Big|_1^3 = 112 \end{aligned}$$

Solution (b).

$$\begin{aligned} \int_2^4 \int_1^3 (40 - 2xy) dx dy &= \int_2^4 \left[\int_1^3 (40 - 2xy) dx \right] dy \\ &= \int_2^4 (40x - x^2y) \Big|_{x=1}^3 dy \\ &= \int_2^4 [(120 - 9y) - (40 - y)] dy \\ &= \int_2^4 (80 - 8y) dy \\ &= (80y - 4y^2) \Big|_2^4 = 112 \quad \blacktriangleleft \end{aligned}$$

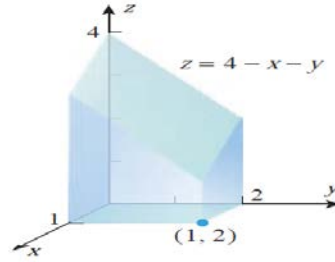
14.1.3 THEOREM (Fubini's Theorem) Let R be the rectangle defined by the inequalities

$$a \leq x \leq b, \quad c \leq y \leq d$$

If $f(x, y)$ is continuous on this rectangle, then

$$\iint_R f(x, y) dA = \int_c^d \int_a^b f(x, y) dx dy = \int_a^b \int_c^d f(x, y) dy dx$$

► **Example 3** Use a double integral to find the volume of the solid that is bounded above by the plane $z = 4 - x - y$ and below by the rectangle $R = [0, 1] \times [0, 2]$ (Figure 14.1.6).



▲ Figure 14.1.6

Solution. The volume is the double integral of $z = 4 - x - y$ over R . Using Theorem 14.1.3, this can be obtained from either of the iterated integrals

$$\int_0^2 \int_0^1 (4 - x - y) dx dy \quad \text{or} \quad \int_0^1 \int_0^2 (4 - x - y) dy dx \quad (8)$$

Using the first of these, we obtain

$$\begin{aligned} V &= \iint_R (4 - x - y) dA = \int_0^2 \int_0^1 (4 - x - y) dx dy \\ &= \int_0^2 \left[4x - \frac{x^2}{2} - xy \right]_{x=0}^1 dy = \int_0^2 \left(\frac{7}{2} - y \right) dy \\ &= \left[\frac{7}{2}y - \frac{y^2}{2} \right]_0^2 = 5 \end{aligned}$$

You can check this result by evaluating the second integral in (8). ◀

► **Example 4** Evaluate the double integral

$$\iint_R y^2 x dA$$

over the rectangle $R = \{(x, y) : -3 \leq x \leq 2, 0 \leq y \leq 1\}$.

Solution. In view of Theorem 14.1.3, the value of the double integral can be obtained by evaluating one of two possible iterated double integrals. We choose to integrate first with

respect to x and then with respect to y .

$$\begin{aligned}\iint_R y^2 x \, dA &= \int_0^1 \int_{-3}^2 y^2 x \, dx \, dy = \int_0^1 \left[\frac{1}{2} y^2 x^2 \right]_{x=-3}^2 dy \\ &= \int_0^1 \left(-\frac{5}{2} y^2 \right) dy = -\frac{5}{6} y^3 \Big|_0^1 = -\frac{5}{6} \quad \blacktriangleleft\end{aligned}$$

The integral in Example 4 can be interpreted as the net signed volume between the rectangle $[-3, 2] \times [0, 1]$ and the surface $z = y^2 x$. That is, it is the volume below $z = y^2 x$ and above $[0, 2] \times [0, 1]$ minus the volume above $z = y^2 x$ and below $[-3, 0] \times [0, 1]$ (Figure 14.1.7).

■ PROPERTIES OF DOUBLE INTEGRALS

$$\iint_R c f(x, y) \, dA = c \iint_R f(x, y) \, dA \quad (c \text{ a constant}) \quad (9)$$

$$\iint_R [f(x, y) + g(x, y)] \, dA = \iint_R f(x, y) \, dA + \iint_R g(x, y) \, dA \quad (10)$$

$$\iint_R [f(x, y) - g(x, y)] \, dA = \iint_R f(x, y) \, dA - \iint_R g(x, y) \, dA \quad (11)$$

It is evident intuitively that if $f(x, y)$ is nonnegative on a region R , then subdividing R into two regions R_1 and R_2 has the effect of subdividing the solid between R and $z = f(x, y)$ into two solids, the sum of whose volumes is the volume of the entire solid (Figure 14.1.8). This suggests the following result, which holds even if f has negative values:

$$\iint_R f(x, y) \, dA = \iint_{R_1} f(x, y) \, dA + \iint_{R_2} f(x, y) \, dA \quad (12)$$

Practice Problem:

Chapter-14.1:

Problems: 3,5,9,10,11,12,29,30,31,32, 35-38

Thank you for your attention!